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Treatment of Chronic Locked Posterior Dislocation of the Shoulder: Is There a Time Limit to Try a Closed Reduction?

A Report of 3 Cases

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Abstract

Case: Three patients with chronic locked posterior shoulder dislocation diagnosed 4 to 10 weeks after injury, each with a reverse Hill-Sachs lesion involving >25% of the humeral head, underwent closed reduction under general anesthesia using the "2 thumbs" technique, followed by immobilization in external rotation.

Conclusion: At a mean 4-year follow-up, all had excellent function without recurrent instability, avascular necrosis, or osteoarthritis. In carefully selected patients, closed reduction remains a safe and effective option beyond the 3-week time frame, when surgery is traditionally recommended.

Neglected locked posterior shoulder dislocation is defined as "chronic" when diagnosed beyond 3 weeks from initial injury[1-3]. This uncommon condition is often associated with high-energy trauma, seizures, or electric shocks. Subtle presentation and insufficient clinical or radiographic assessment may lead to delayed diagnosis[4-6]. Traditionally, closed reduction is considered feasible only within 3 weeks and when the reverse Hill-Sachs lesion (RHSL) involves <25% of the humeral head without associated fractures[3,7]. Beyond these limits, surgical treatment has been the standard approach[1,8,9]. However, prioritizing closed reduction as a first-line, noninvasive treatment option may provide significant benefits for patients. One burning question remains for the surgeon: Until what delay can close reduction be safely and effectively be performed? We present 3 cases of chronic locked posterior shoulder dislocations diagnosed beyond 3 weeks, each involving a RHSL greater than 25%, successfully managed by closed reduction. We describe our reduction technique and provide detailed clinical and radiological outcomes at a minimum 2-year follow-up.

The patients were informed that data concerning their cases would be submitted for publication, and they provided consent.

Case Reports

Case # 1. A 53-year-old right-handed woman presented after 4 weeks of a seizure-related fall. Closed reduction of a posterior right shoulder dislocation was attempted in another hospital, but the patient continued to experience persistent shoulder pain and functional limitation. Clinical examination revealed active elevation limited to 20° and external rotation to -30°. Patient-reported pain Visual Analog Scale (VAS) was 8/10 and Subjective Shoulder Value (SSV) of 30%. No neurological deficits were observed.

Standard radiographs and computed tomography (CT) confirmed a locked posterior dislocation of the right shoulder with a RHSL involving 36% of the humeral head, measured on axial CT slices according to Gerber et al.[10] (Figs. 1-A and 2).

After obtaining informed consent, closed reduction was performed under general anesthesia in the operating room (OR) using the "2 thumbs" (TT) technique. The shoulder was immobilized in 30° abduction and 30° external rotation for 6 weeks, with biweekly follow-up. Rehabilitation first aimed to restore range of motion (ROM), then to strengthen scapulothoracic control and dynamic stabilizers, while avoiding forced internal rotation for 3 months (Fig. 3).

Disclosure: The **Disclosure of Potential Conflicts of Interest** forms are provided with the online version of the article (http://links.lww.com/JBJSCC/C771).

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Keywords posterior shoulder dislocation; chronic locked dislocation; closed reduction; reverse Hill-Sachs lesion; external rotation immobilization

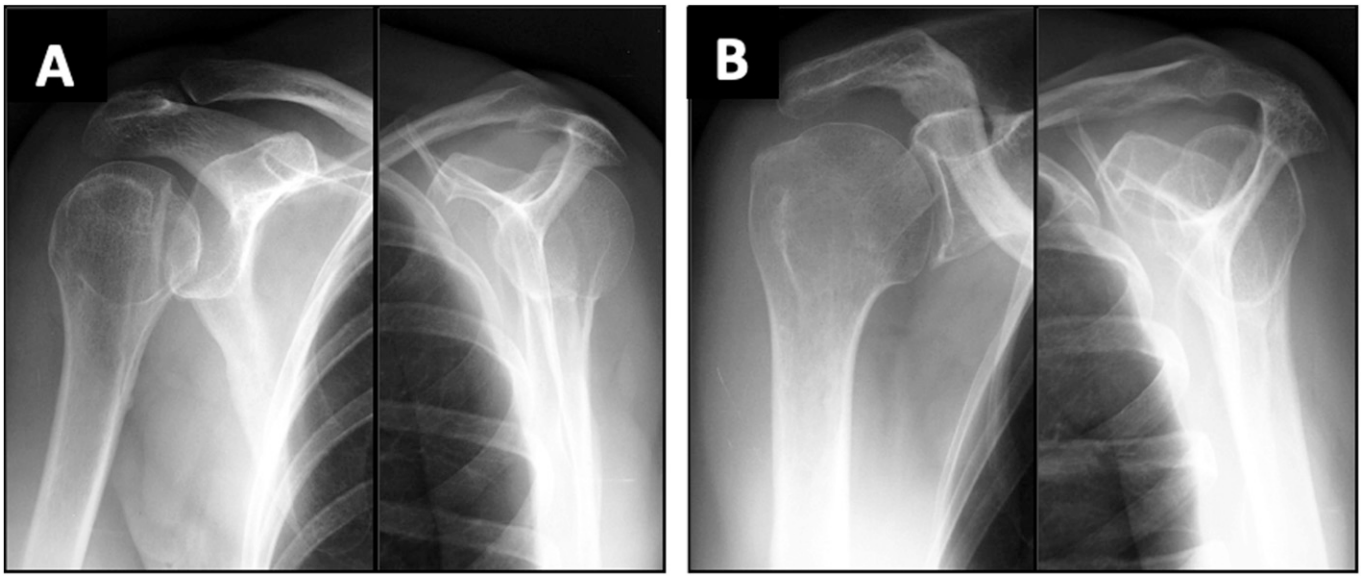


Fig. 1

Radiographic evaluation of Case 1: (**Fig. 1-A**) Prereduction AP view (left) and scapular Y-view (right) radiographs showing a locked posterior shoulder dislocation. **Fig. 1-B** Radiographs at final follow-up demonstrate a well-centered humeral head with no signs of glenohumeral OA. AP = anteroposterior, and OA = osteoarthritis.

At the 5-year follow-up, the patient reported a SSV of 80% and a pain VAS of 2/10. ROM was restored (Fig. 4); (Table I). Follow-up radiographs showed a well-centered humeral head without signs of glenohumeral osteoarthritis (OA) (Fig. 1-B), and no episodes of recurrent dislocation were observed.

Case #2. A 67-year-old right-handed man presented to our clinic 10 weeks after an electrical shock accident at home. He reported chronic moderate right shoulder with nocturnal discomfort and significant functional limitation. Preoperative clinical examination revealed forward elevation limited to 80° and external rotation to −40° (Fig. 5). Patient-reported pain VAS was 3/10, and SSV was 40%. No neurological deficits were noted.

Radiographs and CT scan confirmed a locked posterior dislocation associated with a reverse HSL (Figs. 6-A and 7-A), involving 25% of the anterior humeral head. Magnetic resonance imaging (MRI) revealed a marked hypersignal in the anterior portion of the humeral head on T2-weighted sequences (Fig. 8-A).

Closed reduction under general anesthesia was performed in the OR using the "TT" technique, and intraoperative fluoroscopy confirmed successful reduction (Fig. 6-B). Postoperatively, the shoulder was immobilized for 6 weeks in 30° abduction and 60° external rotation, with biweekly clinical follow-up. Rehabilitation followed the same protocol described for Case 1.

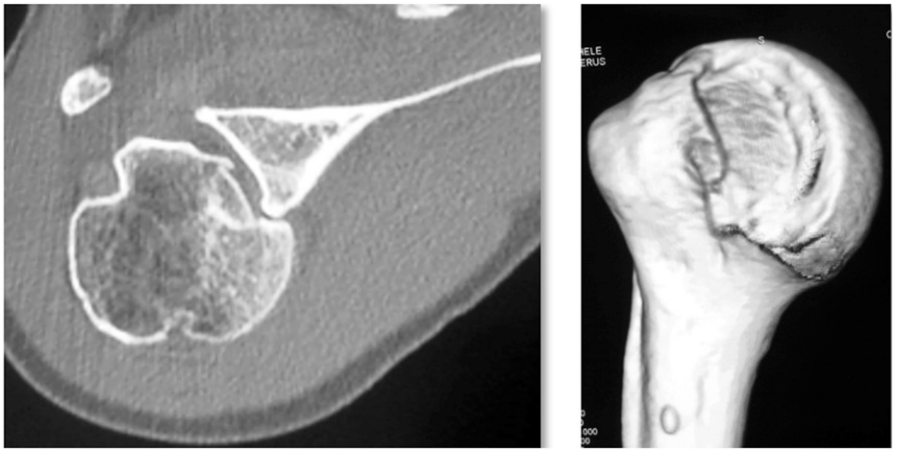


Fig. 2

CT scan of Case 1: Axial CT scan (left) and 3D reconstruction (right) of the right shoulder showing a locked posterior dislocation associated with a prominent anteromedial impaction defect of the humeral head consistent with a RHSL. CT = computed tomography, and RHSL = reverse Hill-Sachs lesion.



Fig. 3

Postoperative immobilization: Clinical image showing shoulder brace in 30° of external rotation and 30° of abduction, maintained for 6 weeks before initiating a structured rehabilitation program.

At 1-year follow-up, MRI revealed complete resolution of the T2 hypersignal (Fig. 8-B).

At the 5-year follow-up, the patient reported complete satisfaction, with SSV of 100%, full active ROM, and no pain (Table I). No complication or recurrence of dislocation was noted. Radiographs and CT scan at the 5-year follow-up confirmed a stable, centered joint with no signs of glenohumeral OA (Fig. 7-B).

Case #3. A 23-year-old right-handed man sustained a traumatic posterior dislocation of the right shoulder during a workplace accident, initially evaluated at another hospital on the day of the injury, where closed reduction was performed. Initial immobilization was performed in internal rotation and followed by a recurrence of the dislocation. The patient was referred to our center 10 weeks after the dislocation and attempted reduction for surgical management.

At presentation, he reported persistent pain and marked functional limitation. Clinical examination revealed active forward elevation of 70° and external rotation to −30°. The pain score was VAS 7/10, and SSV was 10%. Neurological examination was normal.

Standard radiographs and CT imaging confirmed a locked posterior dislocation of the shoulder (Figs. 9-A and 10-A), with a RHSL involving 34% of the anterior portion of the humeral head[10].

Given the clinical and imaging findings, and after obtaining informed consent, a closed reduction was successfully performed under general anesthesia in the OR. No complications were noted, and postoperative imaging confirmed reduction (Figs. 9-B and 10-B)

Postreduction management included 6 weeks of immobilization in 30° abduction and 60° external rotation with biweekly follow-up. Rehabilitation followed the same protocol as in the previous cases.

At the 2-year follow-up, the patient reported complete satisfaction with SSV of 90%. He exhibited a full ROM and was pain-free. Functional Scores were the following: Constant 91, Rowe 80, and Walch-Duplay score 90 (Table I). No complications or recurrence of dislocation were noted. Radiographs and CT images confirmed a centered humeral head and absence of glenohumeral OA (Figs. 9-B and 10-B).

Patient demographic data and outcomes are summarized in Table I.



Fig. 4

Clinical outcome of Case 1: 5-year follow-up demonstrating restored active forward elevation, external, and internal rotation.

TABLE I Patient Demographic Data, Reduction Details, and Clinical Outcomes

	Patient 1	Patient 2	Patient 3
Age (years)	53	67	23
Gender	F	M	M
Side	Right	Right	Right
Etiology	Seizure	Electrical shock	Trauma
% Humeral head defect	36	25	34
Reduction maneuver timing	4 wk	10 wk	8 wk
Type of anesthesia	General	General	General
Period of immobilization	6 wk	6 wk	6 wk
Type of immobilization	30° ER and 30° ABD	60° ER and 30° ABD	60° ER and 30° ABD
Last follow-up	5 yrs	5 yrs	2 yrs
Last follow-up CT scan	Centered humeral head, no OA	Centered humeral head, no OA	Centered humeral head, no OA
Active forward elevation	170	180	180
Active external rotation	50	50	60
Active internal rotation	D12	D12	D12
VAS	2/10	0/10	0/10
SSV	80%	100%	90%
Constant score	79	93	91
Rowe Score	90	100	80
Walch-Duplay Score	90	100	90

ABD = abduction, CT = computed tomography, D12 = level of 12th dorsal vertebra used to describe internal rotation, ER = external rotation, OA = osteoarthritis, SSV = subjective shoulder value, and VAS = pain Visual Analog Scale.

Discussion

Beyond 3 weeks, closed reduction of posterior shoulder dislocation is typically contraindicated due to soft tissue adhesions, and surgery is recommended[11].

In this study, closed reduction succeeded even beyond 3 weeks with RHSL >25%. The "2 Thumbs" technique, developed by the senior author (P.B.), proved effective, followed by 6 weeks of external-rotation immobilization while avoiding internal rotation for 3 months.

Surgical options depend on the defect size of the articular surface. For defects <50%, procedures such as McLaughlin, ORIF, or spherical allografts are used[10,12,13]. For larger defects (>50%), anatomic or semiconstrained shoulder arthroplasty is preferred[4,14-17], though these procedures carry risks of complications and reoperation[16,18].

When chosen, closed reduction should be performed in the OR under general anesthesia with complete muscle relaxation, both of which are essential for success and safety. The

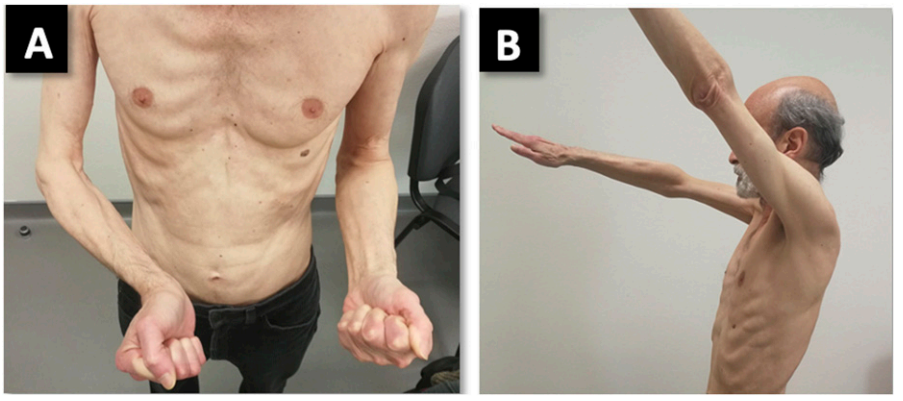


Fig. 5

Preoperative clinical presentation of Case 2, 10 weeks after trauma. **Fig. 5-A** Marked loss of external rotation of the right shoulder. **Fig. 5-**B Limited active anterior forward elevation.

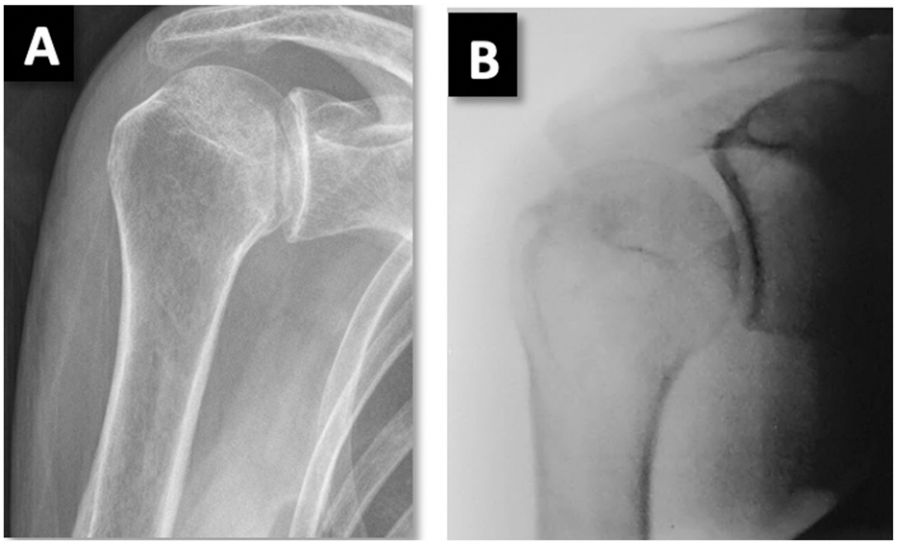


Fig. 6

Radiographic evaluation of Case 2: (**Fig. 6-A**) AP view showing absence of a defined glenohumeral joint space and the presence of a double contour sign, consistent with a locked posterior dislocation. **Fig. 6-B** Intraoperative fluoroscopic image confirming successful reduction. AP = anteroposterior.

patient is placed in lateral decubitus. The surgeon stands behind the shoulder to gently manipulate the humeral head, while the assistant kneels on the table to stabilize the arm (Fig. 11). The TT technique consists of 3 steps: (Fig. 12).

(1) The assistant flexes the elbow to 90° and gradually internally rotates the shoulder to disengage the humeral head from the posterior glenoid rim; 2) the assistant applies a progressive traction to lift the humeral head while the surgeon

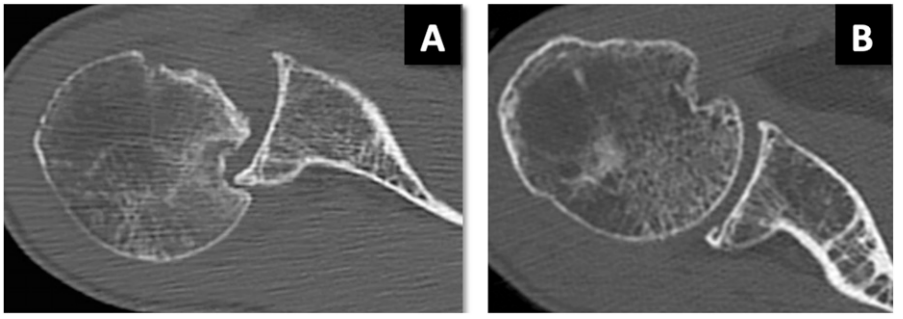


Fig. 7

Axial CT scans of Case 2: (**Fig. 7-A**) Prereduction image showing a locked posterior shoulder dislocation with an associated reverse HSL. **Fig. 7-B** CT scan at 5-year follow-up demonstrating a well-centered humeral head without signs of glenohumeral OA. CT = computed tomography, and OA = osteoarthritis.

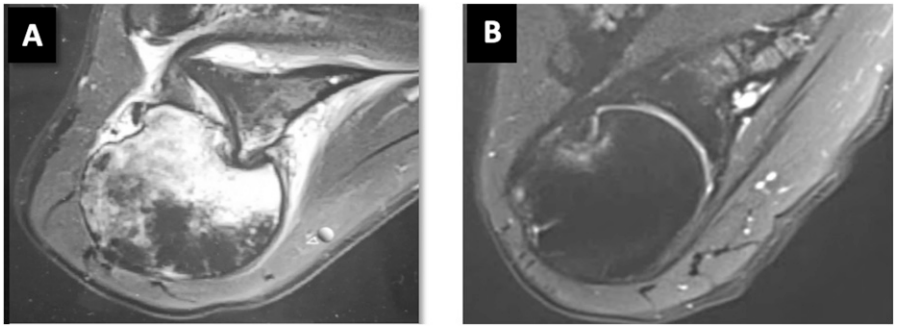


Fig. 8

MRI study of Case 2: Axial T2-weighted sequences showing a marked anterior humeral head hypersignal at diagnosis (**Fig. 8-A**), with complete regression at 1-year follow-up (**Fig. 8-B**). MRI = magnetic resonance imaging.

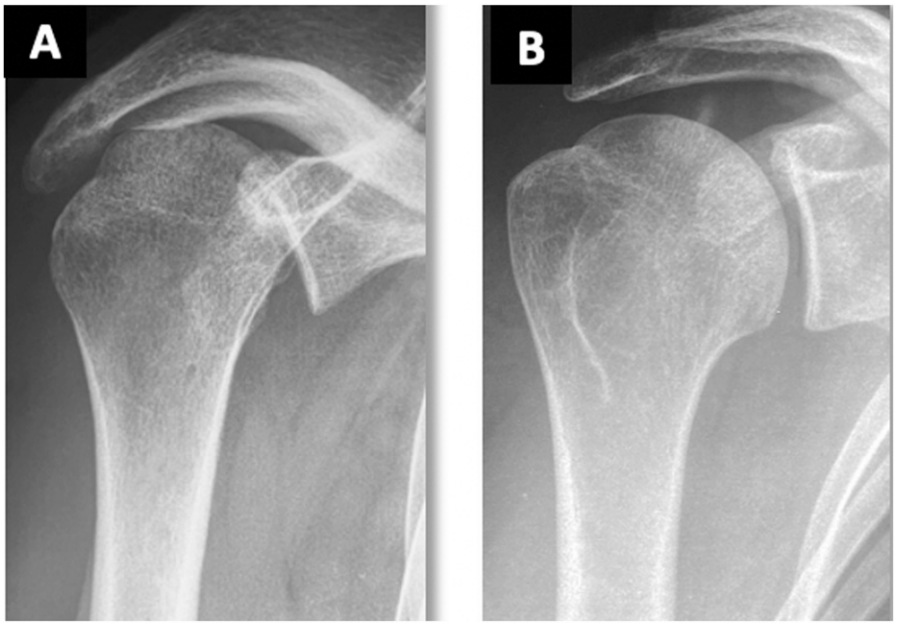


Fig. 9

Radiographic evaluation of Case 3: (**Fig. 9-A**) Prereduction AP views of the right shoulder showing a locked posterior dislocation. **Fig. 9-B** Radiographs at final follow-up demonstrate a centered humeral head with preserved joint space, indirectly indicating maintained cartilage integrity. AP = anteroposterior.

applies a controlled upward and anterior push using both thumbs on the posterior humeral head, guiding it toward the glenoid, and 3) once the humeral head is fully disengaged ("unlocked"), in the third and last step, the assistant is instructed to gently externally rotate the arm. The ability to restore external rotation confirms successful reduction.

Gradual synchronization of manipulation, traction, and rotation ensures smooth reduction with minimal risk. Forced maneuvers must be avoided, and fluoroscopy used to confirm alignment.

Proper immobilization is essential to maintain stability and prevent recurrence. Chronic posterior dislocation elongates posterior soft tissues, creating a capsular pouch that predisposes to redislocation if immobilized in internal rotation. Immobilization in >30° of external rotation promotes capsular healing and fibrous filling of the humeral defect, with the degree of external rotation increased in more chronic cases to address greater posterior soft-tissue elongation and deformation[1,19-22].

In addition to the risk of posterior redislocation, chronic locked posterior dislocations treated with closed reduction may expose patients to stiffness, humeral head avascular necrosis (AVN), and glenohumeral OA. Despite prolonged immobilization, none of our patients developed complications. Functional outcomes were excellent, with full, pain-free ROM and high scores at ≥2-year follow-up (5 years in 2 cases). No cases of AVN were observed. According to the Randelli classification[23],

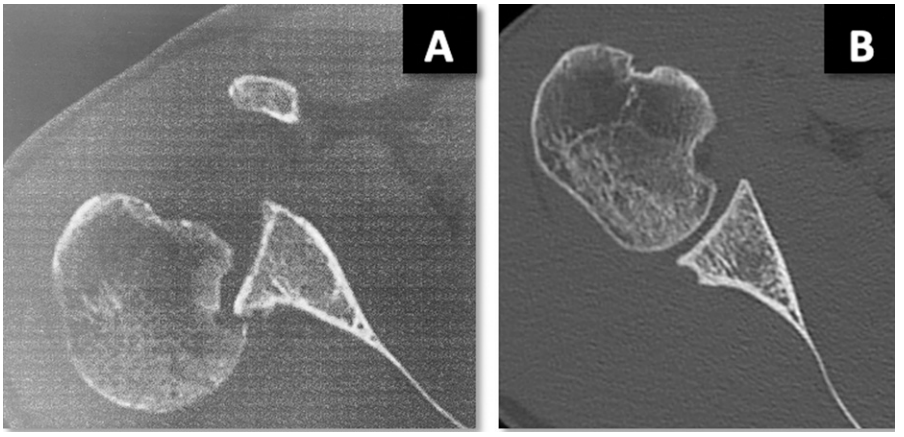


Fig. 10

CT evaluation of Case 3: (**Fig. 10-A**) Axial CT scan before reduction showing a locked posterior dislocation and a reverse HSL. (**Fig. 10-B**) Postreduction CT scan demonstrates a well-centered humeral head without signs of glenohumeral OA. CT = computed tomography, and RHSL = reverse Hill-Sachs lesion.

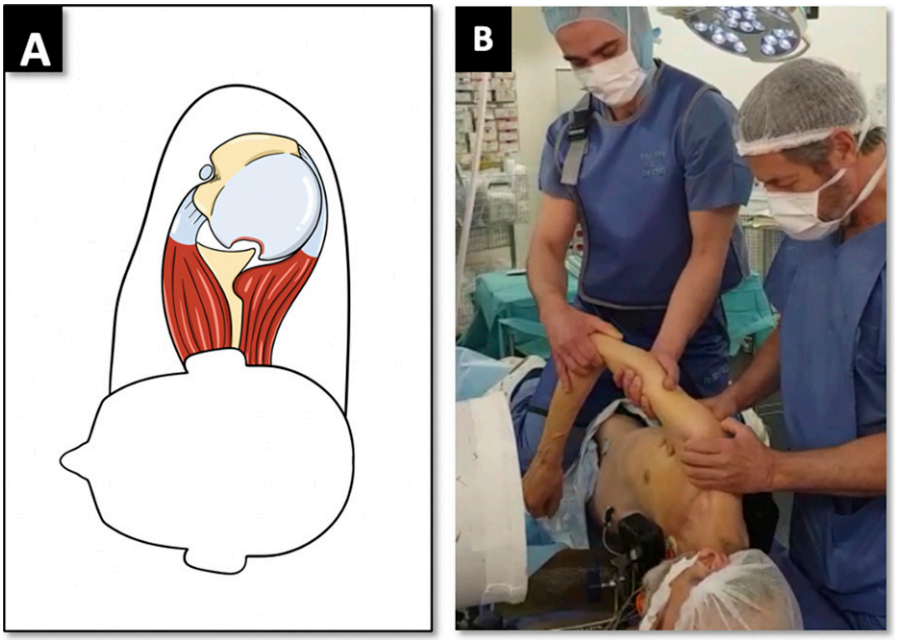


Fig. 11

Fig. 11-A Anatomical illustration showing a locked posterior dislocation and reverse HSL. **Fig. 11-B** Patient positioning and set up: Patient is positioned in lateral decubitus under general anesthesia and complete muscle relaxation. The assistant kneels on the table to stabilize the arm while the surgeon is positioned posteriorly to perform the reduction.

all corresponded to Type-1 posterior fracture dislocation (RHSL <50%), where the risk of necrosis is low[1]. We believe that closed reduction, being less invasive, may better preserve the soft-tissue and vascular integrity of the humeral head compared with open procedures.

At the final follow-up, no signs of glenohumeral OA were observed. These findings align with Hawkins et al., who suggested that intervention within 6 months may allow cartilage preservation[2]. All our cases fell within this time frame, supporting that hypothesis.

To our knowledge, this is the first report describing both the radiological and functional outcomes of closed reduction for chronic locked posterior shoulder dislocation. Further studies with larger series are warranted to validate these results

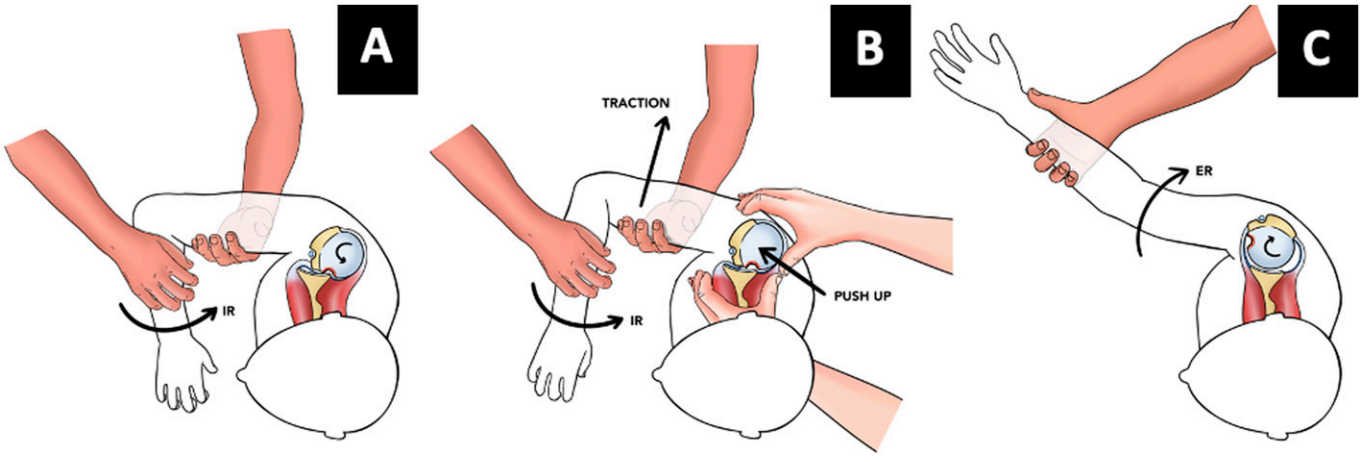


Fig. 12

The TT technique for closed reduction of a chronic locked posterior shoulder dislocation: The maneuver is performed in 3 steps: (**Fig. 12-A**) The assistant progressively increases internal rotation to disengage the humeral head from the glenoid rim; (**Fig. 12-B**) while applying traction, the assistant lifts the arm and the surgeon uses both thumbs to push the humeral head anteriorly, completing disengagement; (**Fig. 12-C**) once the humeral head is unlocked, the assistant externally rotates the arm. Restoration of external rotation confirms a successful reduction. Postreduction, the shoulder is immobilized in >30° of external rotation for 6 weeks to prevent recurrence. TT = 2 thumb.

and to establish this technique as a legitimate alternative to surgery in selected cases.

In summary, closed reduction of neglected chronic locked posterior shoulder dislocation can succeed even beyond 3 weeks, with RHSL >25% but <50% of the articular surface. The reduction using the "TT technique" proved reproducible and safe, with excellent functional recovery and no recurrence, AVN, or OA at ≥2-year follow-up, despite prolonged external-rotation immobilization. ■

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